Nouqta - Improvement Pass (calligraphy, dots, UX, placement test)

You are a senior frontend engineer. The Nouqta MVP works; this pass makes
the calligraphy look like real Naskh, fixes small dots and small text,
simplifies two screens, and adds a placement test at first launch.
All learner text stays in French. Work in the phases below IN ORDER,
finish and self-check each phase, then continue. Do NOT add anything
that is not listed (put ideas in "Suggestions"). For each bug or
decision write the root cause or the reason in one or two sentences.

PHASE 1 - REAL NASKH LETTERS (the most important phase)

Problem

Letters are drawn as uniform "pipe" lines from hand-written centre paths.
They never look like Naskh and several are wrong (Yaa, Waaw, Miim, Qaaf,
Kaaf). Fix by DISPLAYING the real Naskh outline from the Amiri font
(already used by the app, SIL Open Font License, free) and keeping the
centre lines only for pen movement, the ant and validation.

Design

1. Fonts: self-host `public/fonts/Amiri-Regular.ttf` (and keep the Google
   link as fallback). Add `opentype.js`.
2. New module `src/lib/fontGlyphs.ts`:
   - loads the font once;
     `getGlyphOutline(letter, form)` returns SVG
     path data fitted to the 200x200 glyph box, baseline at the glyph's
     `baseline` value, glyph filling about 75 percent of the box width.
   - forms: isolated = the letter; initial = letter + U+200D; medial =
     U+200D + letter + U+200D; final = U+200D +

letter. If opentype.js does
not shape correctly, use the Arabic
Presentation Forms-B code points of
the font instead.

- SPLIT the outline into contours: small
contours (bounding box under 22
percent of the glyph height) are DOTS, the rest
is the BODY. Return
`{ bodyPath, dots: [{ cx, cy, path }] }`.
- verify the number of dot contours equals the
expected dot count
for all 28 letters (Alif 0, Baa 1, Taa 2, Thaa 3,
Jiim 1, Haa 0, Khaa
1, Daal 0, Dhaal 1, Raa 0, Zaay 1, Siin 0, Shiin 3,
Saad 0, Daad 1,
Taa-fort 0, Dhaa-fort 1, Ayn 0, Ghayn 1, Faa 1,
Qaaf 2, Kaaf 0, Laam 0,
Miim 0, Nuun 1, Haa-souffle 0, Waaw 0, Yaa 2).
Print mismatches.

3. Drawing the letter:
    - Voir animation: show the BODY outline filled
    with the ink colour,
    revealed through an SVG `<mask>` whose
    white stroke follows the centre
    line (stroke-dashoffset driven by the existing
    single `progress`
    value, mask stroke width about 1.3 x the
    glyph's maximum thickness).

The ant walks along the centre line. At progress = 1 remove the mask.

- Dots: the real dot contours from the font pop into place one by one, carried by the ant (existing behaviour).
- Tracing: the guide shown to the learner is the font body in light grey (20 percent in light mode); the corridor check still uses the centre line.
- Lists, cards, quizzes, album, Home: use the font glyph as text (Amiri) or the outline path, no monoline.

4. Safety: each letter form gets `renderMode: 'font' | 'monoline'`. Default 'monoline' until its alignment is validated (see 5). Never show a font letter whose centre line is misaligned: the learner would trace one shape and be judged against another.
5. Alignment check (automatic, in /dev/selftest and in /dev/glyphs): draw the body path and the centre line with stroke width 44 on an offscreen canvas and measure (a) the percentage of centre-line samples that lie INSIDE the font body, (b) the percentage of body pixels covered by the

dilated centre line. A letter is 'aligne' when (a) >= 95 percent and
(b) >= 97 percent; set `renderMode: 'font'` only then. Print the
percentages for all 28 letters.

6. Editor `/dev/glyphs`: the reference overlay now shows the font body
outline (toggle 35 percent / 100 percent). Add a button "Suggerer la
ligne centrale" that computes a centre line from the outline
(rasterise, thin to a skeleton, trace from the right-most endpoint,
simplify, fit curves) and loads it for editing. The owner then adjusts
points and validates each letter. Never auto-validate.
7. Forms (Formes button): use the same font method for initial, medial
and final forms, with the keyword highlighting.

Acceptance

- With every letter still on 'monoline', nothing changes visually.
- Switching one aligned letter to 'font' shows real Naskh with thick/thin
strokes in Voir, tracing guide, lists and quizzes.
- The alignment table (28 rows) is printed.

PHASE 2 - DOTS ARE TOO SMALL

Measured: a dot is a diamond with a diagonal of about 12.6 units on a
200-unit glyph (about 12 px on the owner's phone), smaller in the game
cards. Fix:

- Display size: dot diagonal at least 22 units in the lesson canvas (when
  using font dots, scale each dot contour 1.3 x about its centre, check
  clearance again).
- Tap targets for dot steps: 48 px minimum (invisible, around each target).
- Game option cards ("Chasse aux points"): render each card at least 110
  px wide, dots at the same minimum size, cards at least 96 px high.
- "Place les points" canvas: at least 85 percent of the card width; hint
  text 14 px minimum.
- Keep the clearance rule (distance from stroke >= half stroke width +
  half dot diagonal + 6 units); fix any letter that now violates it.

PHASE 3 - QUIZ FIXES

1. "Lecture voyelle": the prompt "quel est le son << ba >> ?" shows the
   transliteration next to each option, which gives the answer away. Two
   question kinds only, never mixed:
   - Hear and choose: speaker button; options are the 3 syllable glyphs
     (no transliteration).
   - Read and choose: show the syllable glyph; options are 3 transliterations
     as text (no glyphs).
     Source: the question builder in src/data/lettersAndFamilies.ts (around
     line 2198) and the sheet that renders it in SheetsAndModals.tsx.
2. Chasse aux points: make `buildDotHuntRounds` return an empty list instead
   of falling back to Baa/Taa/Thaa when fewer than 2 learned letters have
   dots (the UI already switches to Memoire).
3. Album dates: show "29 oct." instead of "2026-10-29".

PHASE 4 - TEXT SIZE, SIMPLER SCREENS, ERRORS

1. Minimum text: nothing under 13 px (the mono labels such as COUVERTURE,
   HORS GUIDE and the game hints are too small). Add a setting "Taille du
   texte" (Normal / Grand / Tres grand = 100 / 115 / 130 percent root size)
   in Reglages; test every screen at 130 percent without overlap.
2. Home (Parcours): keep the hero, the "Continuer" button, the path and the
   3 missions. Move the weekly goal, the "Demain" teaser and the vowels
   card into one collapsible block "Plus" (closed by default, state
   remembered).
3. Moi: split into two segments at the top, "Progres" (profile, ring, album,
   badges, certificate) and "Reglages" (teacher, plan, settings, help).
4. Errors that are silent today:
   - microphone denied or recording failed: show a clear message and block
     the send. REMOVE the placeholder audio string 'NOUQTA_VOICE_30S' in
     useNouqtaState.ts and SheetsAndModals.tsx

so no fake audio can reach
the teacher;

- storage full or write failed (dataService save function swallows
errors): show a persistent banner "Stockage plein : certaines donnees
ne sont plus enregistrees" with a button to delete old photos.

5. Add a "Signaler un probleme" link in Reglages (opens a prefilled
WhatsApp or mail message) and, at the end of each letter, a one-tap
rating (3 faces) saved with the letter id and date.

PHASE 5 - PLACEMENT TEST AT FIRST LAUNCH

Flow

1. After sign-in or sign-up (first time only), show "Ou en es-tu ?" with two
big cards:
   - "Je debute" -> existing flow (tour, then Alif).
   - "Je connais un peu l'arabe" -> placement test.
   Kids accounts: "Je debute" is preselected; the test is still available.

2. Test intro: "2 minutes, 12 questions, sans stress". Buttons: "Commencer" and "Passer".
3. Questions: family by family (F1 to F7), 3 questions per family, reusing the existing generators (hear -> pick the letter; letter -> pick the name; how many dots). Each question has a "Je ne sais pas" button (counts as wrong so nobody guesses). A family is passed with 2 of 3 correct. Stop after 2 consecutive failed families or after 12 questions.
4. Result screen: "Tu peux commencer a : Famille X, lettre Y" with the ant. Letters of passed families become 'acquis': all 5 steps marked done, `source: 'placement'`, stars 1, review box 2 (the spaced review will confirm them in the next days). The unlock rule keeps working (it relies on completed families), so nothing else changes.
5. Plan limits: if the learner is on the free plan and the result is beyond F1, say "Tu es pret pour la famille X" and open the unlock sheet; do not skip the paywall.

6. "Refaire le test" in Reglages. Teacher view shows "Place par test"
   next to those letters.
7. Store `placement: { takenAt, resultFamilyId, answers[] }` in the state.

Acceptance

- A new learner choosing "Je debute" sees no change.
- Simulated learner answering all correctly: F1 to F7 marked acquis in
  order; learner failing F1: stays at Alif.
- A letter marked 'acquis' never unlocks a later family out of order.

MANDATORY SELF-CHECK BEFORE YOU SAY "DONE"

Print PASS/FAIL tables with measured values (extend /dev/selftest):

1. Dot contour counts for all 28 letters match the expected table.
2. Alignment percentages (a) and (b) for all 28 letters, and which are
   `font` vs `monoline`.

3. Dot display size >= 22 units; clearance audit for all 28 letters;
   tap targets >= 48 px on dot steps and game cards.
4. Vowel quiz: 50 generated questions, none shows transliteration and glyph
   in the same option.
5. `buildDotHuntRounds` with 1 dotted letter returns [] (no leak).
6. No text under 13 px anywhere (scan computed styles on all screens);
   130 percent text size: no overlap on Home, Lesson, Moi at 360 px.
7. Microphone denied: no audio is sent; storage failure: banner shown.
8. Placement tests from Phase 5 (all correct, fail at F1, free plan).
9. `npm run lint` and `npm run build` pass; print the output.
   End with: (a) the tables, (b) anything you could NOT verify, stated
   honestly, (c) a short "Suggestions" list (the interactive dictionary and
   stories are NOT part of this pass).